

Response to ISMAR 2026 Papers - submission 1352

AdaptVR: Leveraging Reinforcement Learning and LLMs for Personalized Cybersickness

Mitigation and Reasoning in VR

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We would like to thank all the reviewers for their time and effort in reviewing our work. We found the reviews very helpful and believe these comments will help us improve the content and readability of the manuscript. We have addressed all the comments and suggestions raised by the reviewers.

I. RESPONSES TO THE COMMENTS FROM REVIEWER-1

Concern 1: A key concern is that the RL policy is not trained against comfort directly, but against a proxy reward produced by a sickness classifier trained on self-reported FMS labels. This means the agent is only as reliable as that detector: any label noise, bias, or calibration error can become part of the optimisation target, so lower predicted sickness does not necessarily mean better real comfort.

Response: We thank the reviewer for this comment. In our work, the Transformer classifier is not used as a direct measure of comfort, but only as a real-time cybersickness severity estimator for shaping the RL reward. To reduce the risk of optimizing an unreliable proxy, we first validated the classifier using 10-fold cross-validation, where it achieved 95% overall accuracy and strong per-class F1 scores across cybersickness severity levels. This strong performance suggests that the reward signal is based on a reliable severity predictor rather than a highly noisy or poorly calibrated detector. Moreover, the learned policy was evaluated independently through the user study. Participants provided real-time cybersickness ratings using the FMS scale, and, as reported in Tables 2–4, AdaptVR significantly reduced cybersickness compared with the no-mitigation and static-mitigation conditions. Therefore, the lower cybersickness observed in our evaluation reflects participants’ actual reported comfort levels, not merely lower predicted sickness from the classifier. These results indicate that, in our experimental setting, the Transformer-based reward signal was sufficiently aligned with user-reported cybersickness and exhibited no substantial bias or calibration error affecting the learned policy.

Concern 2: The problem is compounded by a locomotion mismatch in the pipeline. The detector was built from VRWalking data collected during real physical walking, but AdaptVR was evaluated with standing users using controllerbased locomotion, and those two settings can produce different eye and head movement patterns. As a result, the reward signal may not transfer cleanly across movement styles.

Response: We thank the reviewer for this comment. The VRWalking dataset was collected in a small room-scale maze simulation, whereas our custom-built maze simulation is significantly larger. Traversing it physically would require a large tracked space and could introduce physical fatigue as an additional confound. We therefore used controller-based locomotion, a practical and widely adopted VR locomotion method also used in prior cybersickness studies [12, 25, 39], which produces the same core cybersickness-inducing dynamics (e.g., optic flow, turning, etc.) that the detector was trained to recognize. In future work, we plan to conduct an additional real-walking user study in a larger open space, such as an auditorium. We will add this information to our revised version.

Concern 3: The user-experience claims are stronger than the evidence shown. The paper argues that mitigation has little effect on immersion, but the task required users to collect coins, so visibility, awareness, and navigation performance are part of the experience as well as sickness reduction. Because the paper gives little objective task-performance analysis, it is hard to judge the real trade-off between comfort and usability, especially since field-of-view restriction is already known to reduce peripheral scene access.

Response: We thank the reviewer for this comment. Consistent with prior cybersickness mitigation studies [37, 39], we evaluated perceived immersion/users’ immersive experience (UIX) using similar 5-point Likert-scale questions and qualitative feedback after participants completed the maze navigation and coin-collection task. Thus, participants rated AdaptVR after experiencing the relevant visibility, awareness, navigation, and task-interaction demands. As shown in Figure 10, approximately 90% of participants agreed that AdaptVR reduced cybersickness, preserved engagement, and improved overall VR experience compared with static mitigation. Participant feedback, including from P7 and P32, further indicated that adaptive aperture and blur adjustments felt smoother and less visually restrictive than fixed DFOV/DGB settings, as discussed in Section 7. This makes clear that our immersion claim follows the subjective evaluation protocol commonly used in prior cybersickness mitigation work. We also stated in Section 7 that future work will use standardized presence questionnaires to better assess presence, engagement, and immersion.

Concern 4: The comparison is also somewhat limited: DFOV and DGB are reasonable baselines, but they are not the strongest or least intrusive alternatives available. More recent rest-frame style designs aim to reduce discomfort while preserving more of the scene, so the current comparison does not yet show that the RL approach is better than the best available alternatives.

Response: We thank the reviewer for this comment. Our proposed AdaptVR framework dynamically triggers mitigation with accurate intensity based on real-time user state and context. In contrast, even recent rest-frame techniques (e.g., peripheral teleportation [47] and postural-alignment [15]) remain static, follow a one-size-fits-all design, and are applied with fixed or rule-based parameters that do not adapt to real-time user context. Furthermore, our mitigation engine can integrate any mitigation technique with adjustable intensity as noted in Section 3.3. In this work, we validated AdaptVR using DFOV and DGB because they are among the most widely used visual cybersickness mitigation techniques. In future work, we will consider more mitigation techniques as stated in Section 8.

Concern 5: The personalisation method is fairly simple: it mostly responds to what users directly ask for, rather than learning deeper changes in comfort or tolerance over time.

Response: We thank the reviewer for this comment. As described in Section 3.5, user adjustments update the preference vector z , which then adjusts the reward weights $(\lambda_m, \lambda_\Delta)$ and triggers a short PPO fine-tuning step. Thus, the policy incorporates user preferences within the session rather than only applying immediate runtime changes, supporting iterative preference adaptation through repeated user feedback and policy updates. However, long-term personalization across multiple sessions remains an important direction, which we will add to Section 8 of the revised manuscript.

Concern 6: The paper does not clearly show what the LLM adds beyond explanation and control, because most of the comfort gains seem to come from the RL policy and human-in-the loop updates.

Response: We thank the reviewer for this comment. In our work, the LLM is not intended as a direct cybersickness-reduction mechanism but serves as a reasoning engine that supports communication between the user and the RL agent. Specifically, the LLM explains the reasons behind flagged cybersickness, enables natural-language control of mitigation, and captures user preferences that feed into HIL adaptation.

Concern 7: The real-world transfer story is still early: domain randomisation can help, but RL research generally does not treat it as enough on its own, so broader real-world testing is still needed.

Response: We thank the reviewer for this comment. We used domain randomization to improve policy generalization during RL training by exposing the policy to variations in maze layout, motion dynamics, visual flow, and lighting. The trained policy was then validated through our user study, where

AdaptVR significantly outperformed baselines (Tables 2–4). As acknowledged in Section 8, future work will extend training to multiple environments and evaluate AdaptVR in diverse VR contexts beyond the maze simulation.

Concern 8: Name collision: “AdaptVR” has already been used (Sherif, Barakat & Hamdy, 2024 ICCA, dental training).

Response: We thank the reviewer for this comment. The framework will be renamed in the revised manuscript.

II. RESPONSES TO THE COMMENTS FROM REVIEWER-2

Concern 1: The description of the conditions and experimental procedure in Section 6.1.3 requires clarification. While the authors state that the six conditions (including Condition 2’s subconditions, DFOV and DGB) were counterbalanced, the use of sequential language such as “Initially,” “Next,” and “Finally” suggests a fixed order of exposure.

Response: We thank the reviewer for pointing out this ambiguity. The sequential wording in Section 6.1.3, such as “Initially,” “Next,” and “Finally,” was intended only to describe the order in which the conditions are introduced in the manuscript, not the actual exposure order used in the user study. In the experiment, Conditions 1–4 were counterbalanced across participants to reduce order effects. However, the DFOV and DGB subconditions within Condition 2 were presented in randomized order to minimize possible ordering effects between these two static mitigation baselines. The only exception was Condition 5, AdaptVR-HIL, which was necessarily conducted last, on a separate day, because it required the data and user feedback collected during Condition 4 to fine-tune the RL policy via offline PPO updates. We will revise Section 6.1.3 to remove the sequential phrasing and explicitly clarify the scope of counterbalancing, the randomized ordering of the DFOV/DGB subconditions, and the rationale for evaluating AdaptVR-HIL separately after the non-HIL conditions.

Concern 2: More importantly, a significant procedural conflict exists: Condition 5 is dependent on the data and feedback collected during Condition 4 to update its RL policy. This dependency implies that true counterbalancing was only possible for the first five conditions, with the AdaptVR (HIL) condition consistently occurring last, and tested on a separate day. This introduces substantial confounding variables, such as varying washout periods and cybersickness abatement caused by adaptation, which may turn the results for Hypothesis 3 unreliable.

Response: We thank the reviewer for this important concern. Any human-in-the-loop personalization study requires a prior session from which to learn user preferences, so the dependency of Condition 5 on Condition 4 data is structural rather than a design oversight. Regarding the washout and abatement concerns, prior work has shown that cybersickness symptoms can persist from hours up to several days after VR exposure [<https://doi.org/10.1007/s10055-016-0285-9>, <https://doi.org/10.3357/ASEM.2394.2009>], which makes a clean washout or abatement-induced floor effect unlikely. Moreover, a few participants reported severe cybersickness after experiencing Conditions 1–4, and some of them still reported feeling cybersickness in Condition 5 on the following day but felt better while experiencing AdaptVR-HIL. Post-task Likert item S5 (Fig. 10) further showed $\sim 90\%$ agreement that AdaptVR-HIL was preferred over conventional methods. This indicates that AdaptVR-HIL was actively mitigating ongoing discomfort rather than benefiting from a confound-induced effect. We will add participant feedback to Section 7 in the revised manuscript.

Concern 3: Regarding the initial five conditions, it is unlikely that a brief rest period is sufficient to eliminate residual discomfort and prevent carry-over effects.

Response: We thank the reviewer for this comment. As described in Section 6.1.3, rest between conditions was self-paced rather than fixed, with participants continuing only after they reported feeling ready, so no one began a new condition while still discomforted. Moreover, conditions 1–4 were also fully counterbalanced, the standard control for carry-over effects in repeated-measures designs, distributing any residual influence uniformly across conditions rather than biasing any single one. We will clarify both points in the revised Section 6.1.3.

Concern 4: The authors did not report the SSQ distribution of their conditions, arguing that the metric is not real-time. However, SSQ is a more comprehensive questionnaire and is more widely used by researchers. This data is essential for readers to evaluate and crosscompare these findings against existing literature.

Response: We thank the reviewer for this comment. Our work targets the measurement and mitigation of cybersickness *during* immersion, which is why we adopt FMS as the primary measurement, since it is widely used in state-of-the-art real-time cybersickness prediction and mitigation research [27, 37, 39]. We therefore report FMS results explicitly for all conditions. In line with prior work, we additionally report pre- and post-study SSQ data collected before and after the study, as described in Section 6.2.

Concern 5: The parameters for the DFOV and DGB techniques (e.g., FOV mask size or Gaussian

kernel size) were not disclosed. Given that average FMS scores indicate both techniques have similar efficacy, with DGB appearing slightly superior, it is unclear under what logic the RL agent would elect to use DFOV, considering its potentially lower mitigation effect and the objective of preserving peripheral visual content.

Response: We thank the reviewer for this important comment. We will add the specific DFOV and DGB parameter values in the revised Section 3.3. Regarding technique selection, as discussed in Section 3.3, the RL policy selects context-dependent mitigation rather than a single “stronger” one because the effectiveness of each technique depends on the type of motion: DFOV reduces discomfort during rapid rotations [19], while DGB attenuates optic-flow conflicts during forward motion [11]. The HIL preference vector z further biases selection toward each user’s individual tolerance, so the agent does not default to the marginally stronger technique on aggregate.

Concern 6: Minor: no CPU/GPU models reported; “GPT-40” typo; Figure 3 (plain text) might be better as plain text.

Response: We thank the reviewer for identifying these issues. We will address all minor points in the revised manuscript.

III. RESPONSES TO THE COMMENTS FROM REVIEWER-3

Concern 1: Some implementation details could be reported more explicitly, such as the Unity version, ML-Agents version, package/plugin versions, rendering pipeline settings, and integration details between Unity, the RL policy, the cybersickness detection model, and the LLM dialogue engine. Since the paper presents a real-time adaptive VR framework, these details would help readers better assess and reproduce the system.

Response: We thank the reviewer for this comment. In our framework, we used Unity 6 LTS (Universal Render Pipeline), ML-Agents Toolkit 4.0.0, Python 3.10, PyTorch 2.3, Tobii XR SDK 4.x, OpenXR 1.14.3, and XR Interaction Toolkit 3.1.2. The LLM dialogue is implemented via the GPT-4o mini API. The open-source release already contains the full training and inference pipeline along with PPO hyperparameters. We will include these details in Section 4 of the revised version.

Concern 2: The related work is generally comprehensive, and the comparison table is helpful. However, the structure could be made more consistent. Some related-work subsections clearly explain the remaining gap and how AdaptVR differs from prior work, while others mostly

summarize previous studies. Adding one or two concluding sentences at the end of each subsection would help clarify what is novel or different about AdaptVR relative to prior detection-only, static mitigation, rule-based mitigation, RL, and LLM-based VR systems.

Response: We thank the reviewer for this comment. Each major related-work subsection already concludes with a gap statement and differentiation from AdaptVR (e.g., end of Cybersickness Detection, Cybersickness Mitigation, and RL/LLMs in VR). We will strengthen this in the revision by adding concluding sentences that explicitly contrast AdaptVR with prior detection-only, static, rule-based, RL, and LLM-based VR systems for consistent novelty positioning across Section 2.

Concern 3: I also think the LLM and human-in-the-loop personalization components could be evaluated more deeply. The current results support the overall usefulness of the framework, but the evidence for how much the LLM dialogue and HIL adaptation independently contribute could be clearer. This does not substantially weaken the paper, but a more detailed analysis would strengthen the claims.

Response: We thank the reviewer for this comment. Our study design already isolates these components, as described in Section 6.1.3 and reflected in Tables 2–4 with the corresponding statistical analysis. Specifically, Condition 3 (RL only) evaluates adaptive mitigation without the LLM, Condition 4 (AdaptVR) adds the LLM dialogue layer without HIL fine-tuning, and Condition 5 (AdaptVR-HIL) adds personalization. The FMS-mitigation gap between RL and AdaptVR reflects the LLM/dialogue contribution, while the gap between AdaptVR and AdaptVR-HIL isolates personalization. The LLM is not intended as a direct cybersickness-reduction mechanism (please refer to our response to R1-C6 for further details on the LLM’s role). We will clarify this in Section 6.1.3 and make the contributions of the LLM and HIL components more explicit in the revised manuscript.

Once again, we appreciate the time and effort of the reviewers. We can ensure that the reviewers’ recommendations will be reflected in the revised/camera-ready version of the manuscript.

Thankfully

Authors of 1352